

Tuning Melt Strength and Processability of Polyolefins by Addition of a Functionalized Additive Designed via the TEMPO-Driven Thiol-ene Reaction

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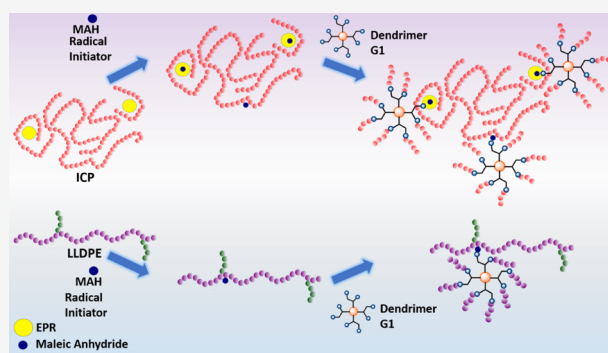


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ABSTRACT: Thiol-ene chemistry with TEMPO as an initiator has been utilized to synthesize a first-generation dendrimer having eight peripheral hydroxyl groups. Long chain branched impact copolymer polypropylene (ICP) and linear low-density polyethylene (LLDPE) were prepared by a postreactor modification process by a melt-grafting dendrimer onto a polyolefin chain using peroxide and maleic anhydride as a co-agent. Long chain branching in ICP controlled the chain scission initiated by peroxide by increasing entanglement, while for LLDPE, it controlled the excessive crosslinking because of peroxide. Neat and modified ICP and LLDPE samples were analyzed by FT-IR, TGA, DSC, HT-GPC, MFI, and parallel-plate rheometry as well as for their melt strength. The presence of long chain branching was mainly confirmed by time–temperature superposition (TTS) analysis obtained using a parallel-plate rheometer. In comparison to neat polymers, modified ICP and LLDPE samples showed higher values of Arrhenius flow activation energy. In the case of ICP, higher processability was observed for a dendrimer-grafted sample, while modified LLDPE samples exhibited higher melt strength as compared to neat polymers. Van Gurp–Palmen plots showed a decrease in phase angle as a function of angular frequency for modified ICP and LLDPE samples, which also confirmed their thermorheological complex behavior.



INTRODUCTION

In the past few years, dendrimers and hyperbranched polymers (HBPs) have attracted considerable attention for their unique structure and properties. They are globular macromolecular structures that contain a central core having multiple branches terminating with reactive functional groups, which can be modified further to obtain desirable properties. While dendrimers have a fractal structure, uniformly grown branches, specific molecular weight, and known surface reactivity, HBPs have a relatively imperfect and polydisperse architecture due to uncontrolled growth reaction. Dendrimers form an important class of molecules that have applications ranging from drug delivery, nanoreactors, catalysis, molecular devices, analytical chemistry, sensors, etc.^{1,2} Moreover, they have proven to be excellent candidates for modifying the properties of polymers remarkably such as improving the mechanical and rheological properties^{3,4} and rate of crystallization⁵ or increasing miscibility by acting as a compatibilizer in the matrix.⁶

Dendrimers have played a key role in the modification and functionalization of polyolefins that is quite a challenging task from an industrial point of view. Polyethylene (PE) and polypropylene (PP) are the most widely used thermoplastics in the world due to their attractive chemical and mechanical properties. PP offers low cost as well as good moisture,

chemical, and electrical resistance. Linear low-density polyethylene (LLDPE) too has appreciable tear and puncture resistance and possesses high elongation at break and tensile strength. However, the poor chemical reactivity because of the hydrophobic nature of polyolefins makes it difficult to further modify or improve their properties. There are several examples in the literature where dendrimers have been extensively used to modify the properties of polyolefins. Boogh *et al.* functionalized two commercially available hyperbranched dendritic polymers named Boltorn-H20 and Boltorn-H30 and analyzed them as property modifiers for PP-glass fiber composites.⁷ While Boltorn-H20 effectively promoted the interfacial adhesion between PP and glass fibers, Boltorn H-30 acted as a processing aid by smoothening the surface of composites after injection molding or extrusion owing to the low viscosity of HBP. In another study, Liu and Zhao observed that 5% of hyperbranched polyamide ester in isotactic PP

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resulted in remarkable acceleration in the crystallization rate in PP.⁸ The fractal globular shape of HBP resulted in reduced viscosity and chain entanglement that eventually promoted the movement of polymeric chains toward the growth of nuclei. Hong *et al.* observed that functionalized Boltorn H-30 acted as a lubricant in the LLDPE extrusion process by migrating toward the outer surface of the polymer to aid in overall processability.⁹ Kuang *et al.* observed that blends of poly-(amidoamine) dendrimers and ultrahigh-molecular-weight PE displayed an effective reduction in melt viscosity as well as improved crystallization behavior and tensile strength.¹⁰

The chemical modification in PP and LLDPE has been explored over the years to modify their properties to further increase the scope of applications. PP has poor impact properties due to which several reinforcing agents or fillers need to be added in the matrix. The inert nature of the PP matrix leads to weak interfacial adhesion and, therefore, poor dispersion of additives. In addition to this, efforts have been made to enhance the processability of polyolefins during industrial processes to increase the rate of production that can lead to huge cost savings for industries. The impact properties of PP can be increased by copolymerizing it with ethylene to obtain impact copolymer polypropylene (ICP), which is a biphasic polymer where ethylene propylene rubber (EPR) particles are dispersed in the PP matrix. Though ICP has higher impact strength than homopolymer PP, both possess poor melt strength. The narrow processing window in PP leads to lack of a rubbery plateau during transition from a solid crystalline form to a melt state during melting processing of PP. This is a direct indication of low entanglement or crosslinking in polymer chains.^{11–13} The most attractive method of increasing entanglement and improving the melt properties is by introducing long chain branching in the polymer chains. The prime objective of introducing long branched chains onto the linear polymer backbone is to tune its melt strength and processability. With improved melt properties, the end-use application profile can be highly extended to important polymer processes such as coating, spinning, sheet casting, and film blowing.

Over the years, postreactor modification has proven to be a great technique to incorporate additives or fillers to meet the requirements for industrial and commercial purposes. The poor reactivity between hydrophobic polyolefins and hydrophilic additives can be improved by several monomers including maleic anhydride,¹⁴ glycidyl methacrylate,¹⁵ acrylic acid,¹⁶ and others that can be grafted onto the backbone of base polymers such as LLDPE, LDPE, PP, etc. These functional groups impart polarity to the nonpolar polymer interface that further increases its reactivity and interfacial adhesion with other polymer matrices or polar organic compounds. Functionalized polyolefins are largely used in packaging, lamination, printing, coating, and formation of compatibilized blends with polymers such as polyamides and polyesters.¹⁷ The most commonly used reagent for the modification of polyolefin is maleic anhydride (MAH), which acts as a reactive co-agent and links polyolefins and polar materials through covalent bonds. The modification of polyolefins has been commonly performed by radical polymerization which results in chain scission and increasing branching, grafting, or crosslinking. It is also possible to alter the viscosity of the extruded product by initiating degradation and introducing long chain branching in the polyolefin backbone. Along with processability, incorporating branching

has also proven to affect the polymer's crystallinity, density, melt strength, environmental stress crack resistance, and melt fracture in extrusion processes.¹⁸ The reactive compounds having a low molecular weight are often employed for initiating crosslinking reactions and formation of copolymers. For example, peroxides are added in a few grades of PE and PP as radical initiators for controlled degradation. The melt blending of peroxides with polyolefins results in simultaneous scission and crosslinking reactions. The dominant reaction during extrusion depends on several factors, such as the amount of peroxide, reaction temperature, reaction time, etc. Postreactor modification of PP melt with peroxides generally proceeds with β -scission, leading to a decrease in the viscosity of the matrix, whereas, in the case of linear PE, the reaction of peroxide with the melt state primarily progresses with branching or crosslinking, leading to an increase in the viscosity of the system.^{5,19} During the reactive extrusion process, functionalized additives help in controlling the β -scission that competes with the grafting reaction in polypropylene and help improve melt properties in LLDPE.^{20–22}

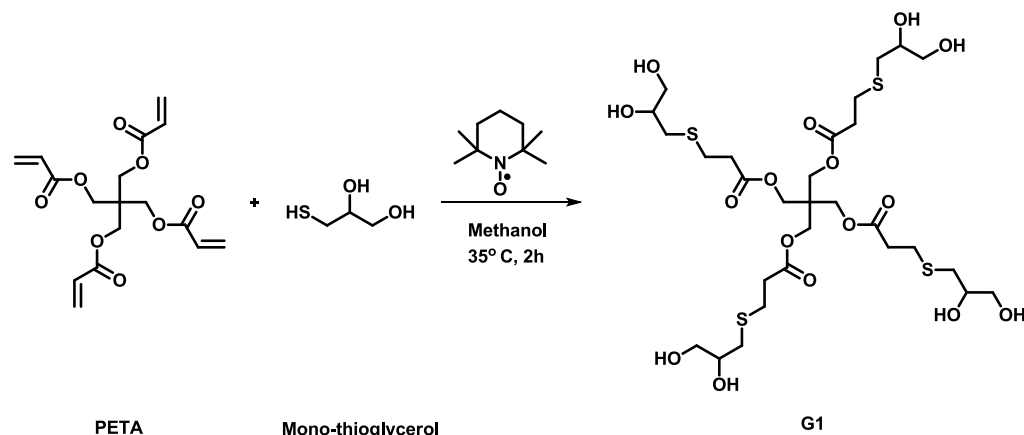
Taking into consideration the synthetic procedures utilized for the preparation of dendrimers, the conventional methods often include multiple protection/deprotection steps and complicated purification of the final product.^{23,24} In contrast, thiol-ene chemistry has emerged as a popular tool to synthesize dendrimers due to its ease of performing a reaction and orthogonal nature. The regioselectivity of the reaction offers good yield of the desired product without the tedious purification procedure.^{25–27} Hafeez *et al.* demonstrated the formation of polymeric microspheres using the thiol-ene reaction, establishing TEMPO as an initiator operating under ambient conditions and using benign solvents such as water.²⁸ The oxidation–reduction process initiated in TEMPO results in the formation of a thiolate anion that adds to the C=C bond that leads to the corresponding product. Combining the flexibility and simplicity of thiol-ene chemistry with the high efficiency of TEMPO can prove to be a breakthrough for the synthesis of dendrimers and other macromolecular structures.²⁹

Postreactor chemical modification or functionalization of polyolefins has been achieved using industrially scalable methods by melt extrusion³⁰ and solid-state pulverization.^{31–33} However, our present study of the chemical modification of polyolefins is one of the very few to do using dendrimers. In this work, the synthesis of generation-1 dendrimer having terminal eight hydroxyl groups (G1) has been carried out by using TEMPO as an initiator for thiol-ene chemistry. It offered an improved methodology for synthesis involving no complex purification procedure and 100% yield. G1 was further incorporated in the impact copolymer grade of polypropylene (ICP) and LLDPE chains through reactive compatibilization using MAH. The prime idea was to explore the efficiency of the synthesized dendrimer as an additive in polyolefins. Luperex-101 was used to generate radicals for the grafting of G1 via MAH as a reactive co-agent to introduce long chain branching in the polyolefin chains. Its effect has been extensively studied through thermal, chemical, and rheological methods of characterization.

■ EXPERIMENTAL SECTION

Materials. Impact copolymer (ICP) polypropylene-grade PP 3120MA consisting of 8–10% ethylene propylene rubber

Scheme 1. Preparation of a G1 Dendrimer from PETA as the Central Core Using the TEMPO-Initiated Thiol-ene Reaction



and polyethylene-grade LLDPE-010F18S were supplied by Indian Oil Refinery, Panipat, having MFI values of 12 g/10 min (230 °C and 2.16 kg) and 0.99 g/10 min (190 °C and 2.16 kg), respectively. Pentaerythritol tetraacrylate (PETA) (TCI Chemicals) was purified by hexane/ethyl acetate for the synthesis purpose. 1-Monothioglycerol, the free radical initiator Luperox-101 (Luperox), and the thiol-ene reaction initiator (2,2,6,6-tetramethylpiperidin-1-yl)oxyl or (2,2,6,6-tetramethylpiperidin-1-yl)oxidanyl [TEMPO] were supplied by Sigma-Aldrich and were used as such. Maleic anhydride (Thermo Scientific) was used as a reactive co-agent during melt blending of ICP and LLDPE samples. Among solvents, methanol (HPLC grade) was used as a medium for the thiol-ene reaction and hexane, ethyl acetate, and diethyl ether (all from Thermo Fisher Scientific) were used for purification purposes.

Synthesis of the Dendrimer G1. PETA was used as the central core for the desired dendrimer synthesis and was purified by column chromatography using 30% hexane/ethyl acetate as an eluent system prior to reaction. Five grams of PETA (14.2 mmol) and 5 mL (56.8 mmol) of 1-monothioglycerol were reacted in the presence of 1.42 mmol of TEMPO in 65 mL of methanol as shown in Scheme 1. The reaction medium was kept for stirring at 35 °C for 2 h. The solvent mixture was then evaporated to around 1/5th of its quantity followed by precipitation in around 450 mL of cold diethyl ether to get rid of colored impurity to obtain a clear viscous oil in 100% yield. The purified dendrimer was characterized by ^1H NMR, ^{13}C NMR, FT-IR, and TGA.

^1H NMR (400 MHz, CD_3OD), δ ppm: 4.07 (s, 8H), 3.62 (m, 4H), 3.42–3.38 (m, 8H), 2.68 (t, J = 6.8 Hz, 8H), 2.5–2.62 (m, 12 H), 2.45 (dd, J = 12 Hz, 6.8 Hz, 4H).

^{13}C NMR, δ ppm: 171.1, 71.5, 61.5, 62.1, 41.2, 34.8, 34.3, 27.2.

Preparation of Modified Samples of ICP and LLDPE with G1. Modified samples of LLDPE with G1 were prepared using a twin screw extruder LZ120/Vs by Lab Tech Engineering Company at temperature settings of 160–170–175–180–185 °C and a 100 rpm screw speed. The extrudate was water-cooled and pelletized. Luperox concentration and MAH loading for LLDPE samples were 100 ppm and 3%, respectively. G1 was dissolved in 30 mL of ethanol to ensure homogeneity in the compositions and was added in 0.5, 1, and 2% samples. All the samples were manually mixed before melt blending.

ICP-G1 samples were prepared by melt mixing using a Haake Rheomix 19/25 Rheodrive 16 mixer at 180 °C with a rotor speed of 50 rpm. Antioxidants AO 1010 (500 ppm), AO 168 (1000 ppm), and calcium stearate (500 ppm) were added in all the compositions. The preheat and mixing times for all the compositions melt-blended in the rheomix were 3 and 5 min, respectively. For modified ICP samples, Luperox concentration and MAH loading were fixed as 50 ppm and 3%, respectively, while G1 was added in 1 and 2% samples. All the prepared samples and detailed formulations are listed in Table 1.

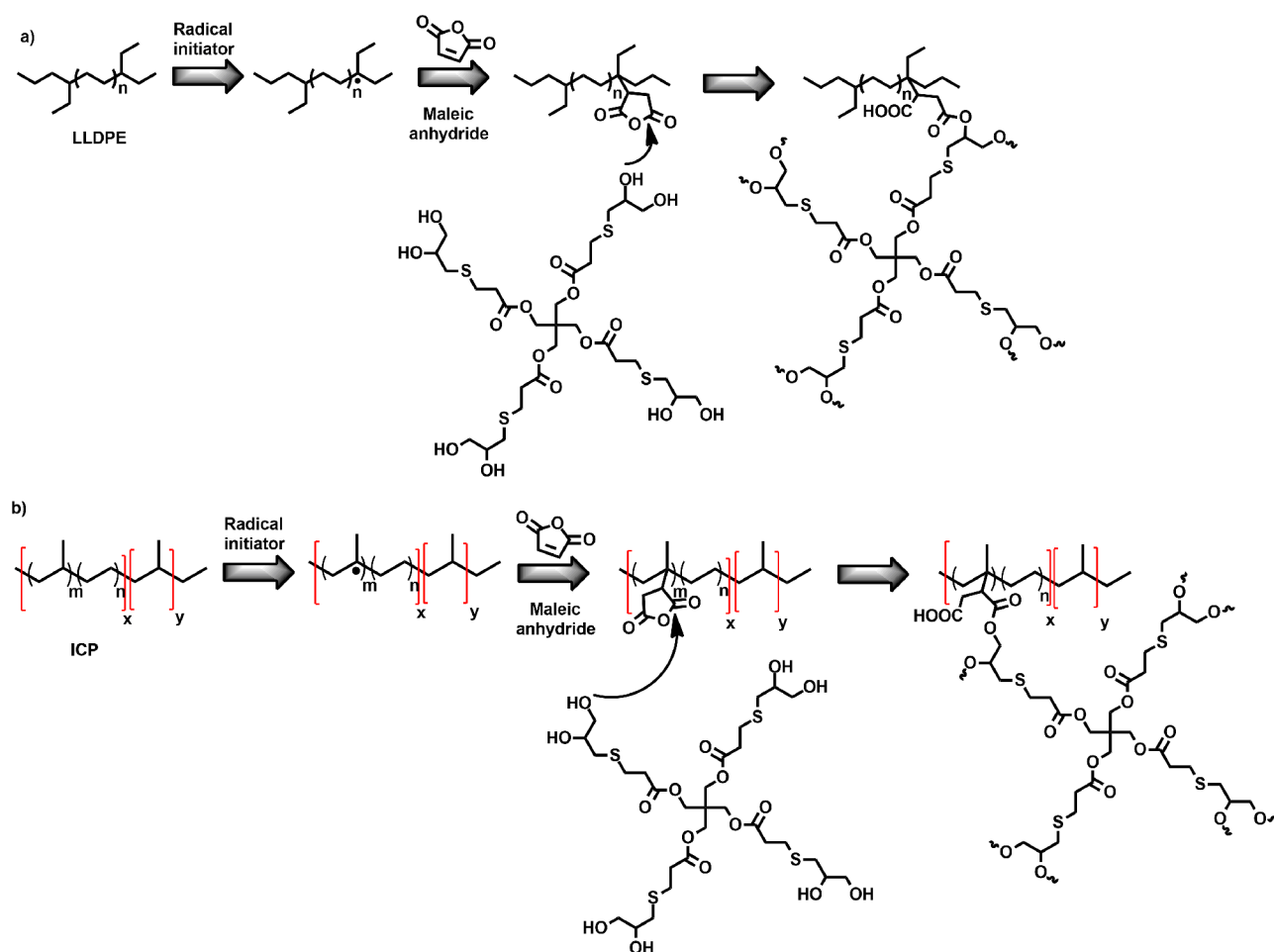
Table 1. ICP and LLDPE Samples Modified with G1

Composition	Abbreviation	Luperox (ppm)	MAH (%)	G1 (%)
Neat ICP	Neat ICP			
ICP + Luperox	ICP + Lup 50	50		
ICP + Luperox + MAH	ICP + Lup 50 + 3% MAH	50	3	
ICP + Luperox + MAH + 1%G1	ICP + Lup 50 + 3% MAH + 1% G1	50	3	1
ICP + Luperox + MAH + 2% G1	ICP + Lup 50 + 3% MAH + 2% G1	50	3	2
Neat LLDPE	Neat LLDPE			
LLDPE + Luperox	LLDPE + Lup 100	100		
LLDPE + Luperox + MAH	LLDPE + Lup 100 + 3% MAH	100	3	
LLDPE + Luperox + MAH + 0.5% G1	LLDPE + Lup 100 + 3% MAH + 0.5% G1	100	3	0.5
LLDPE + Luperox + MAH + 1% G1	LLDPE + Lup 100 + 3% MAH + 1% G1	100	3	1
LLDPE + Luperox + MAH + 2% G1	LLDPE + Lup 100 + 3% MAH + 2% G1	100	3	2

Characterization. FT-IR Spectroscopy. FT-IR spectra were recorded using a Shimadzu spectrometer IRPrestige-21 to qualitatively determine the grafting of the dendritic modifier G1 on LLDPE and ICP chains via MAH. The spectra were measured by co-adding 50 scans over a region of 4000–400 cm^{-1} at a spectral resolution of 4 cm^{-1} . The thin films for FT-IR microscopic analysis were prepared by hot-pressing using a Specac CTFM at 150 °C under 5 MPa pressure. The melting time for the thin films was 3 min and their thickness was about 25 μm .

Xylene-Soluble Portion. The xylene-soluble portion was obtained according to the ASTM D5482-17 procedure for ICP samples. A known weight of sample (around 2 g) was dissolved

Scheme 2. Plausible Mechanism of the Grafting of a G1 Dendrimer on (a) LLDPE and (b) ICP Chains through Maleic Anhydride



in 200 mL of xylene and stirred under a continuous nitrogen atmosphere at 200 °C. It was then kept at room temperature for 12 minutes followed by cooling at 25 °C for 30 minutes. The resultant ICP solution was filtered to obtain the xylene-insoluble (XIS) portion, while 100 mL from the filtrate was evaporated to yield the xylene-soluble (XS) portion. Around 400 mL of acetone was added to the remaining 70 mL of filtrate and was stirred for 5 min to obtain rubber as a precipitate. All XS, XIS, and rubber phases obtained were dried in an oven at 70 °C for around 2.5 h and their weight was recorded.

¹³C NMR Spectroscopy. ¹³C NMR spectra were recorded to estimate the change in ethylene (C2) content in the ICP sample with a 1% grafted dendrimer as compared to neat ICP and ICP with maleic anhydride grafted (control sample) using an Agilent 500 MHz DD2 NMR spectrometer with a 10 mm dual probe at 130 °C. The following spectral parameters were set for the analysis: spectral width of 31,250 Hz, pulse of 90°, relaxation delay of 14 s, acquisition time of 1.05 s, and number of scans of 3000.

Thermal Characterizations. Differential scanning calorimetry (DSC) was performed on a Temperature-Modulated Mettler Toledo DSC 1 star instrument under a nitrogen environment. Samples were heated from 25 to 200 °C and held in the molten state for 5 minutes to remove the influence of thermal history. The sample mass was about 5–6 mg and a heating and cooling rate of 10 °C/minute was utilized.

Thermogravimetric analysis (TGA) was conducted on a TA instrument Q 500. It was performed under a nitrogen atmosphere at a heating rate of 10 °C/minute within a temperature range of 25 to 750 °C.

Rheological Characterization. ICP and LLDPE samples were characterized for their melt flow index by a Göttfert MI 4. All the measurements were recorded for 2.16 kg load at temperatures of 230 °C for ICP and 190 °C for LLDPE. The rotational rheological behavior of samples was investigated by conducting a small amplitude oscillatory shear (SAOS) test using a TA-ARES G2 rheometer (using parallel-plate mode) on 25 mm-diameter plate with a 1 mm gap size. Circular samples (1 mm) were obtained by compression molding using a Collin P 400P/M instrument at 210 °C (for ICP) and 180 °C (for LLDPE) and a 50 bar pressure. The tests were performed in the frequency range of 0.628–628 rad s⁻¹ at a fixed strain of 3%, obtained from strain amplitude sweep test at the frequency of 10 rad s⁻¹, to ensure that tests are within the linear viscoelastic range of deformation. The analysis was recorded at 170–180–190–200 °C for ICP and at 160–170–180–190 °C for LLDPE. Zero-shear viscosity, crossover frequency, and activation energy of flow for all the samples of LLDPE and ICP were estimated by evaluation at 160 and 190 °C, respectively.

The melt strength of the samples was also measured under similar temperature conditions for ICP and LLDPE as mentioned above, using a Göttfert Rheotens 71.97 instrument,

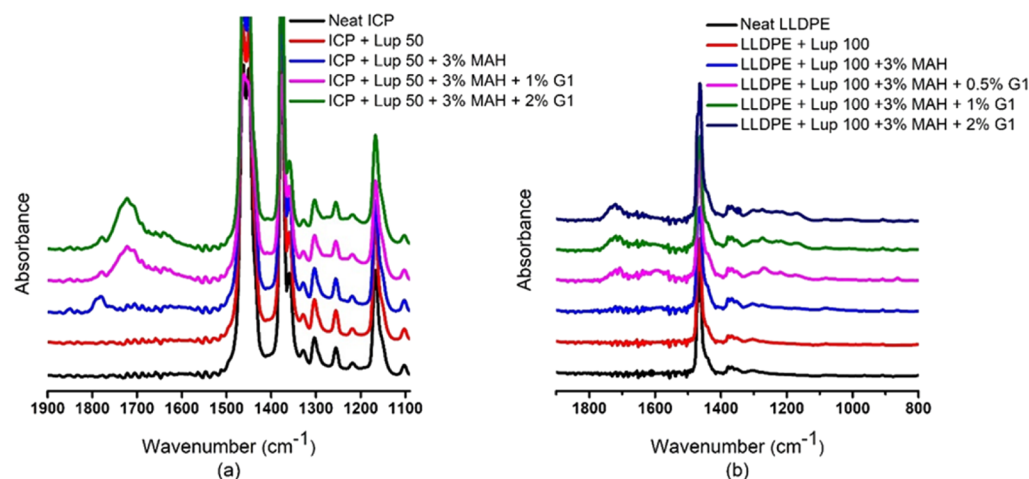


Figure 1. FT-IR spectra for (a) ICP-G1 samples and (b) LLDPE-G1 samples to estimate the amount of grafting of a dendrimer via maleic anhydride.

Table 2. Absorption Band Ratio of ICP and LLDPE Samples Modified with G1

Composition	Absorption band ratio (ester) in ICP-modified samples at 1722 cm ⁻¹ /1167 cm ⁻¹	Absorption band ratio (ester) in LLDPE-modified samples at 1720 cm ⁻¹ /1463 cm ⁻¹
Neat ICP	0.004	
ICP + Lup 50	0.004	
ICP + Lup 50 + 3% MAH	0.03	
ICP + Lup 50 + 3% MAH + 1% G1	0.3	
ICP + Lup 50 + 3% MAH + 2% G1	0.5	
Neat LLDPE		0.02
LLDPE + Lup 100		0.02
LLDPE + Lup 100 + 3% MAH		0.05
LLDPE + Lup 100 + 3% MAH + 0.5% G1		0.07
LLDPE + Lup 100 + 3% MAH + 1% G1		0.08
LLDPE + Lup 100 + 3% MAH + 2% G1		0.08

Germany, in combination with an Advanced Twin Bose Capillary Rheometer by Malvern Instruments, U.K.

High-Temperature Gel Permeation Chromatography (HT-GPC). Gel permeation chromatography (GPC) data were obtained on a Polymer Char model GPC-IR in 1,2,4-trichlorobenzene at 160 °C. A three-column set was employed, which consisted of PL gel 300 × 7.5 mm Agilent columns. The calibration of the columns was performed using polystyrene standards and covering the range for molecular weight from 10³ to 10⁷ g/mol. The calculations were then performed to convert the calibration curve for polystyrene to PP using the Mark–Houwink constants. The sample concentration for LLDPE was 1 mg/mL, while that for ICP was 0.5 mg/mL. The injection volume was 200 μL and the flow rate was maintained at 1 mL/min.

RESULTS AND DISCUSSION

The synthesis of G1 was confirmed by ¹H NMR and ¹³C NMR spectroscopies (given in Figures S1 and S2, respectively). The absence of resonances for the vinyl hydrogen in the 5.8–6.5 ppm region and emergence of triplet around 2.7 ppm confirmed the thioether bond formation. FT-IR spectrum analysis of G1 (Figure S3) showed bands corresponding to –OH stretching vibrations, –CH stretching and bending

vibrations, and –C=O stretching vibrations. Thermogravimetric analysis (TGA) showed the onset of degradation (*T*_{onset}) for G1 to be 277 °C, which was well above the processing window of polyethylene and polypropylene (given in Figure S4). G1 was used as a reactive additive for the modification of ICP and LLDPE in the presence of a radical initiator and maleic anhydride. The detailed formulations studied in this work are discussed in Table 1. The proposed mechanism for reactive compatibilization in LLDPE and ICP and the synthesized dendrimer with hydroxyl terminal groups aided by MAH is given in Scheme 2.

The confirmation of interaction between G1 and ICP/LLDPE has been studied using various techniques. The spectroscopic characterization by FT-IR was carried out to confirm the grafting of G1 on the polyolefin chain through MAH and is shown in Figure 1 for (a) ICP and (b) LLDPE (complete FT-IR spectra shown in Figure S5). The amount of maleic anhydride grafted was determined by the carbonyl index, which was calculated by the ratio of absorption (band height) of the anhydride band with respect to absorption at 1167 cm⁻¹ (characteristic IR frequency of CH₃ groups relative to polypropylene).³⁴ The reference band for polyethylene samples was chosen to be at 1463 cm⁻¹ corresponding to the bending vibration of CH₂.^{35,36} The characteristic bands of

MAH in the case of ICP samples can be observed in the spectrum of ICP + Lup 50 + 3% MAH at 1782 and 1849 cm^{-1} , which confirmed the grafting of MAH. These bands were absent in neat ICP and ICP + Lup 50 compositions. Samples with 1 and 2% G1 depicted a strong band at 1722 cm^{-1} that corresponds to the stretching vibration due to the carbonyl group present in ester (absent in the control sample with MAH grafting), which has been formed by the reaction of the hydroxyl group of the dendrimer and carbonyl carbon of MAH. Consequently, the increase in the absorption band ratio at 1722 cm^{-1} /1167 cm^{-1} to 0.3 in ICP with 1% G1 and up to 0.5 with 2% G1 was observed with the increase in the amount of G1 (as shown in Table 2). Similarly, the carbonyl frequency for polyethylene samples was calculated by the ratio of band heights at 1720 cm^{-1} /1463 cm^{-1} and was found to increase with the addition of G1.

Xylene-soluble test has been performed to separate the amorphous portion (xylene-soluble) from the crystalline portion (xylene-insoluble) part in ICP. The xylene-soluble part consists of both the acetone-soluble portion as well as the acetone-insoluble part (rubber phase). We observed that the rubber content decreased in modified ICP samples as compared to neat ICP, as given in Table 3. This can provide

Table 3. Xylene-Soluble Portion (%), Rubber Portion Obtained (%), and C2 Content (from ^{13}C NMR) Obtained for ICP-G1 Samples

Composition	Xylene-soluble portion (%)	Rubber content (%)	C2 content (^{13}C NMR) (weight %)
Neat ICP	18.6	9.5	14.0
ICP + Lup 50	18.5	7	
ICP + Lup 50 + 3% MAH	18.6	5.5	11.9
ICP + Lup 50 + 3% MAH + 1% G1	18.3	5.1	10.7
ICP + Lup 50 + 3% MAH + 2% G1	18.5	5.1	

an indication that the reaction was taking place in the amorphous portion, and therefore, its weight decreased. However, the total amorphous portion or the xylene soluble value remains unchanged for all the compositions. The possible reason could be that for ICP modified with G1, some portion by mass was added to the acetone-soluble part from the xylene-insoluble part.

Table 4 shows the comparison for grafting measured in terms of carbonyl frequency for all ICP-G1 samples. For samples modified with G1, the ratio of carbonyl frequency for modified ICP with respect to neat ICP was calculated and the corresponding values were compared for xylene-soluble and

xylene-insoluble samples, which confirmed that the grafting of G1 was found to be greater on the xylene-soluble portion. The FT-IR spectra for the same are depicted in Figure 2 (complete range spectra shown in Figure S6).

^{13}C NMR spectroscopy has been a primary mode for analyzing the microstructure of impact co-polymer PP (ICP). The ^{13}C NMR spectra of ICP samples did not show any new peaks that could correspond to branching or grafting. It is often agreed that NMR is not an effective tool for characterizing long chain branching as the chemical shift for a co-monomer having more than six or eight carbon atoms cannot be distinguished from that of long chain branching in the absence of high-resolution studies.^{37,38} However, it was used for determining the overall content of co-monomer ethylene (C2) for three ICP samples. The resonances in the 18–24 ppm region corresponds to the pendant methyl group in polypropylene, while the complete resonances of ethylene combined with polypropylene were observed in the 10–50 ppm region. The integration of signals corresponding to ethylene and polypropylene in ^{13}C NMR was converted to mole fraction and then weight fraction using the method proposed by Kumar *et al.*³⁹ The ethylene contents of neat ICP, control sample, and modified sample with 1% G1 were examined for their C2 content, as shown in Table 3.

As discussed above, grafting on ICP chains has been found to be greater on the amorphous portion by FT-IR. From Table 3, it can be observed that the ethylene content decreased from 14 for neat ICP to 11.6 for the control sample with MAH and further decreased to 10.7 for ICP modified with 1% G1. The decrease in the signals corresponding to ethylene shows that it was consumed in the grafting reaction of MAH with G1 and hence indicates the presence of an external moiety or branching being introduced in the ICP backbone. The ^{13}C NMR spectra for the above-mentioned ICP samples are provided in the Supporting Information (Figure S7).

The addition of Luperox decreased the onset degradation temperature (T_{onset}) for both ICP and LLDPE. Thereafter, the grafting of G1 through MAH improved the thermal stability post addition of Luperox for both ICP and LLDPE (comparison shown in Figure S8). Table 5 shows the impact of enhanced branching on T_{onset} obtained through TGA and thermal parameters like crystallization peak temperature (T_c), melting point (T_m), and enthalpy of fusion (ΔH_m) for ICP and LLDPE samples. A slight increase was observed in T_{onset} in ICP-G1 samples as compared to neat ICP with addition of 1 and 2% additives. There was no significant change observed in T_c , T_m , and ΔH_m for ICP and LLDPE samples (complete thermogram shown in Figure S9).

HT-GPC was performed for all ICP and LLDPE samples along with the rubber phase of ICP-G1 samples, and the

Table 4. Comparison of Measure of the Grafting of G1 in Xylene-Soluble and Xylene-Insoluble ICP-G1 Samples

Composition	Carbonyl frequency for xylene-soluble samples	Ratio of carbonyl frequency = modified ICP/neat ICP	Carbonyl frequency for xylene-insoluble samples	Ratio of carbonyl frequency = modified ICP/neat ICP
Neat ICP	0.13		0.37	
ICP + Lup 50	0.07		0.34	
ICP + Lup 50 + 3% MAH	0.10		0.33	
ICP + Lup 50 + 3% MAH + 1% G1	0.36	2.8	0.45	1.2
ICP + Lup 50 + 3% MAH + 2% G1	0.44	3.5	0.51	1.3

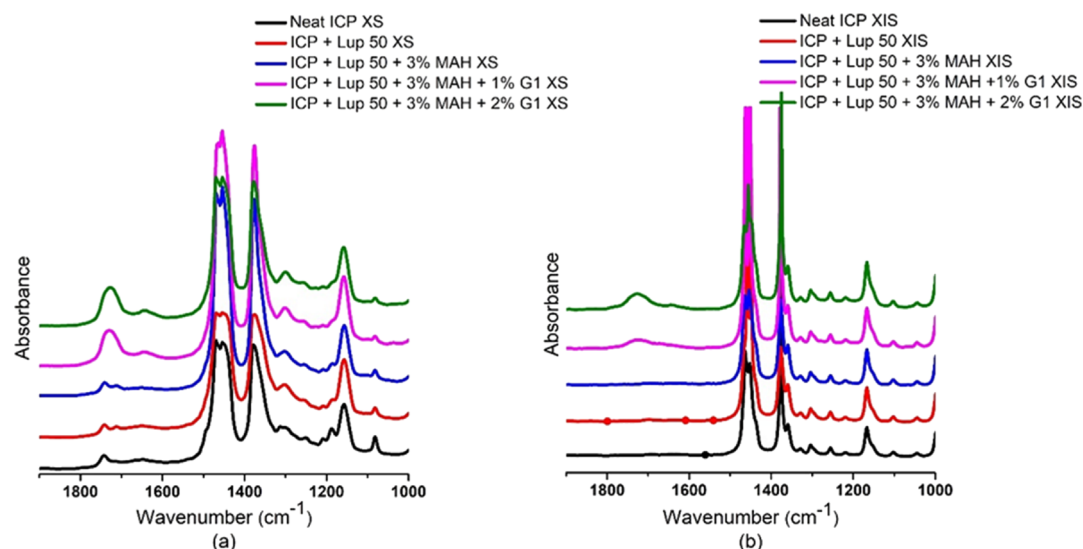


Figure 2. FT-IR spectra of (a) xylene-soluble and (b) xylene-insoluble ICP-G1 samples.

Table 5. Onset Degradation Temperature of ICP and LLDPE Samples Obtained through TGA and DSC Parameters Crystallization Peak Temperature (T_c), Melting Point (T_m), and Enthalpy of Fusion (ΔH_m)

Composition	TGA		DSC	
	T_{onset} ($^{\circ}\text{C}$)	T_c ($^{\circ}\text{C}$)	T_m ($^{\circ}\text{C}$)	ΔH_m (kJ/mol)
Neat ICP	403	116	163	76
ICP + Lup 50	332	116	163	75
ICP + Lup 50 + 3% MAH	423	116	162	73
ICP + Lup 50 + 3% MAH + 1% G1	428	118	163	72
ICP + Lup 50 + 3% MAH + 2% G1	428	117	163	74
Neat LLDPE	457	103	121	41
LLDPE + Lup 100	423	103	121	41
LLDPE + Lup 100 + 3% MAH	452	104	119	40
LLDPE + Lup 100 + 3% MAH + 0.5% G1	448	105	120	41
LLDPE + Lup 100 + 3% MAH + 1% G1	438	104	119	40
LLDPE + Lup 100 + 3% MAH + 2% G1	451	105	120	41

parameters obtained, such as weight-average molecular weight (M_w), number-average molecular weight (M_n), polydispersity index ($\text{Đ} = M_w/M_n$), z-average molecular weight (M_z), and intrinsic viscosity $[\eta]$, are listed in Table 6. The addition of the radical initiator Luperox to neat ICP leads to a decrease in the M_w due to chain degradation as a consequence of β -scission of the tertiary macroradicals. This resulted in a decrease in the intrinsic viscosity and Đ as well. The grafting G1 onto the ICP chain helped in controlling the chain degradation further that caused a slight increase in M_w , Đ , and intrinsic viscosity as compared to the one modified by the radical initiator. It should be noted here that M_w values of modified samples were not higher than that of neat ICP. The probable reason for this could be that chain scission and grafting/branching are occurring simultaneously in the polymer chains, but the impact of β -scission has been reduced due to improved entanglement with the grafting of G1.⁴⁰

Table 6. HT-GPC Parameters for Neat and Modified ICP and LLDPE Samples

Composition	M_w (g/mol)	M_n (g/mol)	M_w/M_n	M_z (g/mol)	$[\eta]$ (dL/g)
Neat ICP	371,000	36,000	10	1,184,000	1.8
ICP + Lup 50	332,000	40,000	8.2	962,000	1.6
ICP + Lup 50 + 3% MAH	333,000	39,600	8.4	988,000	1.7
ICP + Lup 50 + 3% MAH + 1% G1	352,000	39,000	9	1,068,000	1.8
ICP + Lup 50 + 3% MAH + 2% G1	336,000	41,000	8.3	976,000	1.7
Neat ICP rubber	378,000	96,000	3.9	851,000	1.9
ICP + Lup 50 rubber	259,000	88,000	2.9	506,000	1.4
ICP + Lup 50 + 3% MAH rubber	502,000	100,000	5	1,422,000	2.3
ICP + Lup 50 + 3% MAH + 1% G1 rubber	448,000	102,000	4.4	1,353,000	2.1
ICP + Lup 50 + 3% MAH + 2% G1 rubber	490,000	105,000	4.7	1,612,100	2.2
Neat LLDPE	303,200	46,700	6.5	1,377,300	3.6
LLDPE + Lup 100	302,200	45,700	6.6	1,168,800	3.6
LLDPE + Lup 100 + 3% MAH	290,000	49,400	5.9	1,135,500	3.5
LLDPE + Lup 100 + 3% MAH + 0.5% G1	290,100	49,300	5.9	1,136,300	3.5
LLDPE + Lup 100 + 3% MAH + 1% G1	291,600	49,000	5.9	1,167,800	3.5
LLDPE + Lup 100 + 3% MAH + 2% G1	291,200	48,800	6	1,164,200	3.5

HT-GPC results for the rubber phase samples obtained by extraction in acetone from the xylene-soluble part of ICP were more pronounced, and parameters obtained are listed in Table 6. Figure 3 shows a shift in the molar weight distribution (MWD) toward lower molecular weights as a result of chain scission due to addition of the radical initiator in ICP. This was followed by an evident shift in MWD toward higher molecular weights for ICP samples grafted with G1, which is due to crosslinking provided by G1. M_w and $[\eta]$ for modified ICP-G1 samples were even higher than those of neat ICP, which indicates enhanced entanglement in the amorphous portion of the ICP chain by the grafting of G1 (parameters listed in Table

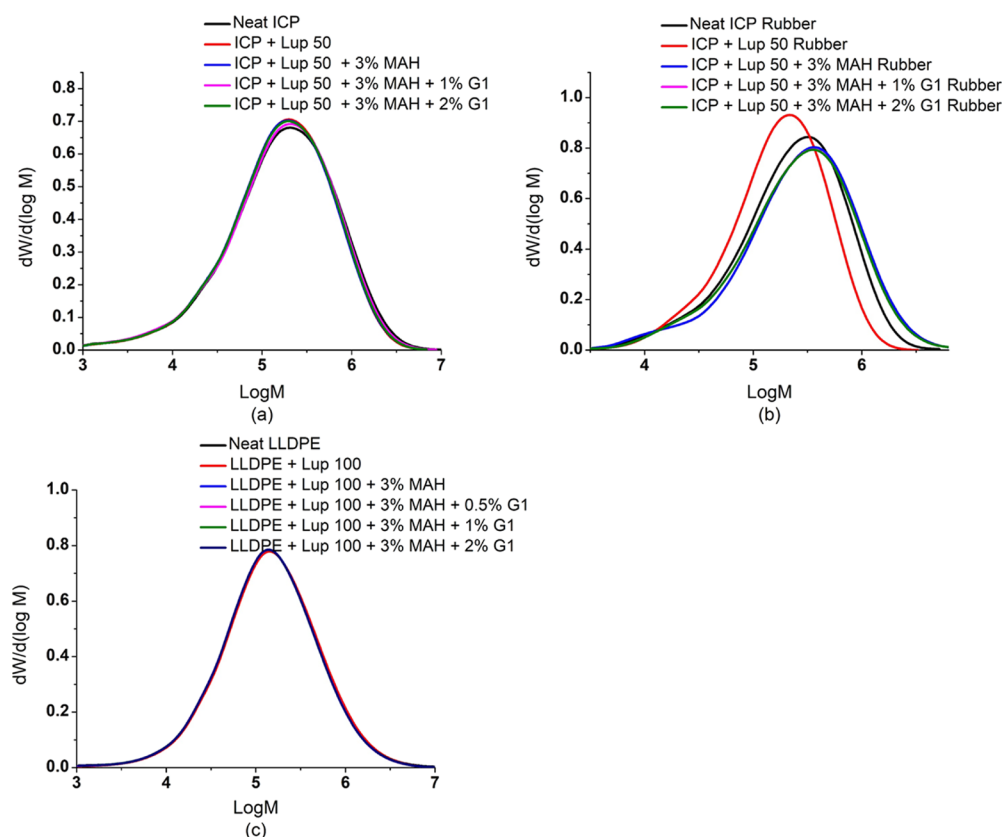


Figure 3. Changes in the molar weight distribution (MWD) for (a) neat ICP and ICP-G1 samples, (b) the rubber portion of neat ICP and ICP-G1 samples, and (c) neat LLDPE and LLDPE-G1 samples.

Table 7. Melt Flow Index (MFI), Melt Strength (MS), Zero Shear Rate Viscosity (η_o), Crossover Modulus (CM), Relaxation Time (λ), and Activation Energy (E_a) in Neat and Modified ICP and LLDPE Samples

Composition	MFI	MS (cN)	η_o (Pa s)	λ (s)	CM (Pa)	E_a (kJ/mol)
Neat ICP	13	1.33	2083	0.07	21,068	35.5
ICP + Lup 50	26	1.07	829	0.03	23,929	33.7
ICP + Lup 50 + 3% MAH	20	0.9	1445	0.04	33,283	31.4
ICP + Lup 50 + 3% MAH + 1% G1	16	1.33	1596	0.05	25,886	62.3
ICP + Lup 50 + 3% MAH + 2% G1	15	1.31	1033	0.04	18,345	25.5
Neat LLDPE	1.03	6.3	9867	0.07	73,943	16.6
LLDPE + Lup 100	0.53	11.4	100,909	0.13	85,808	34.3
LLDPE + Lup 50 + 3% MAH	0.89	8.9	56,817	0.11	105,347	48.2
LLDPE + Lup 50 + 3% MAH + 0.5% G1	0.92	8.6	35,996	0.10	76,845	50.6
LLDPE + Lup 50 + 3% MAH + 1% G1	0.94	8	42,451	0.11	103,947	58.6
LLDPE + Lup 50 + 3% MAH + 2% G1	0.75	9.2	32,986	0.11	117,753	68

6). The increase in $\bar{D}$ for modified ICP-G1 samples as compared to neat ICP was observed, which indicated the presence of both linear and branched chains in the polymer backbone.

GPC analysis was of less significance in LLDPE-G1 samples. There was no change observed in the molecular weight or intrinsic viscosity of the modified samples. Since the determination of GPC parameters depends on the radius of gyration (R_g) of a molecule, it is therefore possible that the apparent molecular weight of the modified samples is much lower than their actual molecular weight because R_g for a branched molecule is much less than that of a linear molecule for the same molecular weight.⁴¹ The effect of grafting G1 on LLDPE chains was extensively studied by rheological analysis

using parallel-plate rheometry and melt strength measurements, which are discussed in the section below.

Rheological Characterization. Melt Flow Index. Melt flow index (MFI) is the measure of ease of flow of the polymer melt under a set of defined conditions. As discussed above, during the modification of polyolefins initiated by the radical initiator, crosslinking is the main side reaction in polyethylene, while polypropylene proceeds primarily via chain scission. As given in Table 7, MFI for neat ICP was 13, which increased to 26 on addition of 50 ppm of Luperox. A low concentration of Luperox was added to polypropylene to control the chain degradation process in ICP. With addition of MAH, the effect of chain scission was reduced, resulting in decreased MFI. This effect became more prominent with grafting of G1 on the ICP chain as MFI decreased further with addition of 1 and 2% G1.

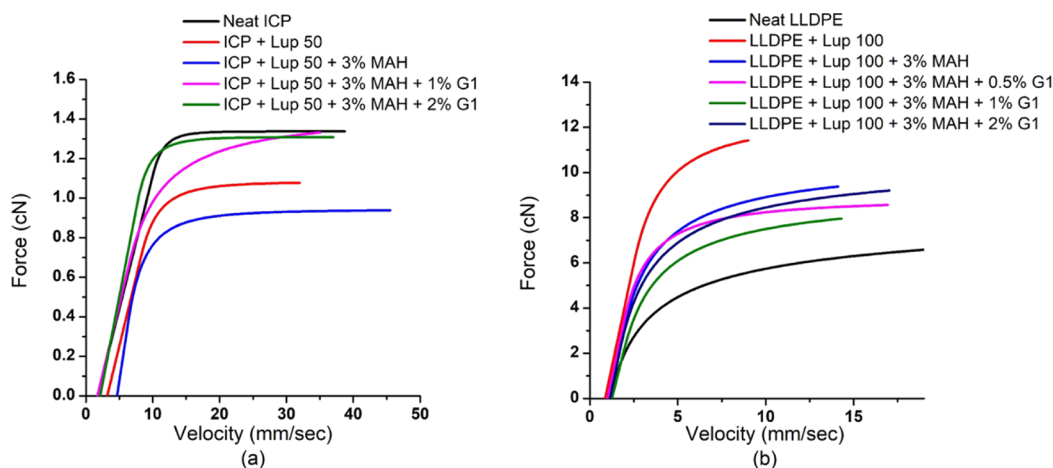


Figure 4. Comparison of melt strength of (a) neat ICP and ICP-G1 samples and (b) neat LLDPE and LLDPE-G1 samples.

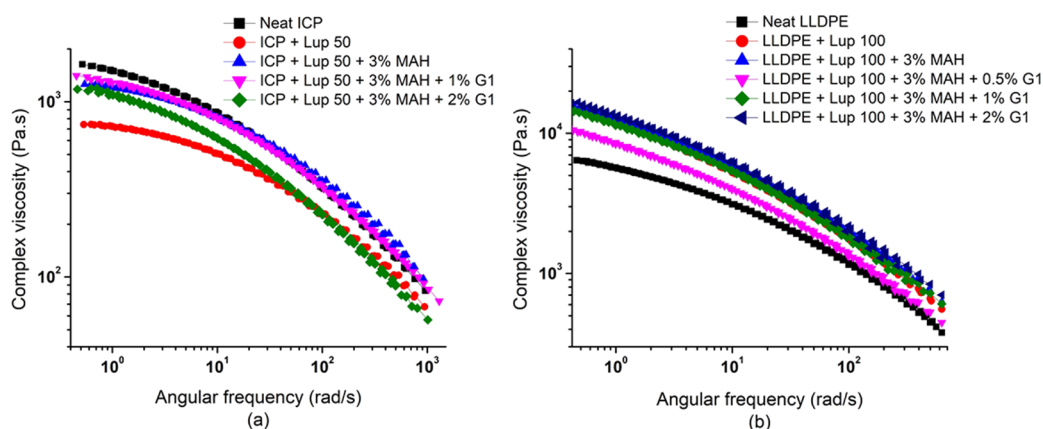


Figure 5. Complex viscosity versus angular frequency plots of (a) neat ICP and ICP-G1 samples and (b) neat LLDPE and LLDPE-G1 samples.

In LLDPE, the addition of 100 ppm of Luperox decreased the MFI by around 50%, indicating enhanced crosslinking. The processability of LLDPE improved with the grafting of MAH on the LLDPE chain. MFI further increased to as close as that of neat LLDPE due to long chain branching introduced with addition of 0.5 and 1% G1. However, with addition of 2% G1, MFI decreased, resulting in reduced processability due to a high degree of crosslinking.

Melt Strength. Melt strength is the measure of resistance of the polymer melt to stretching. It is defined as the maximum tension or force that polymer melt can bear without breaking. Linear polymers such as PP, LLDPE, and HDPE do not possess high melt strength because of poor chain entanglement. Therefore, efforts are made to alter the molecular architecture to increase branching to obtain higher melt strength and extensional viscosity.⁴²

In our studies, the addition of Luperox to ICP increased the chain scission and consequently decreased melt strength (as shown in Table 7). Grafting MAH has often proven to enhance the scission reaction further, which is exactly what has been observed for the control sample of ICP, Luperox, and MAH.⁴³ It can be observed clearly from Figure 4a that melt strength increased with the grafting of 1 and 2% additives as compared to that modified by the radical initiator and MAH. Though the melt strength of neat ICP and that modified with 1% G1 is comparable, the latter has a higher MFI, which indicates enhanced processability for the modified sample.

The introduction of long chain branching in LLDPE has brought appreciable increase in the melt strength as compared to neat LLDPE due to enhanced entanglement by the grafting of G1. The addition of 100 ppm of Luperox showed the highest value of melt strength due to increased crosslinking but had a poor draw ratio, as shown in Figure 4b. It decreased slightly after the grafting of MAH onto the LLDPE chain. With modification by 0.5 and 1% G1, a noteworthy balance of MFI and melt strength was observed. The modified samples possessed processability close to neat LLDPE with increased magnitude of melt strength (as shown in Table 7). Melt strength further increased for LLDPE modified with 2% G1 but had a low MFI or poor processability as compared to neat LLDPE.

Parallel-Plate Rheology. Rheology is defined as the study of flow or deformation of matter and utilized to determine viscoelastic properties of a material. It is often considered a reliable method as compared to the traditional methods such as GPC and NMR as the latter are not always able to provide quantitative information regarding long chain branching.⁴⁴ Rheological parameters such as η_0 , E_a , and relaxation time (λ) are sensitive to the degree of branching and hence provide valuable information regarding the molecular structure and entanglement in the polymer. These entities obtained from the SAOS experiment are temperature-dependent, and their corresponding values mentioned in Table 7 are measured at

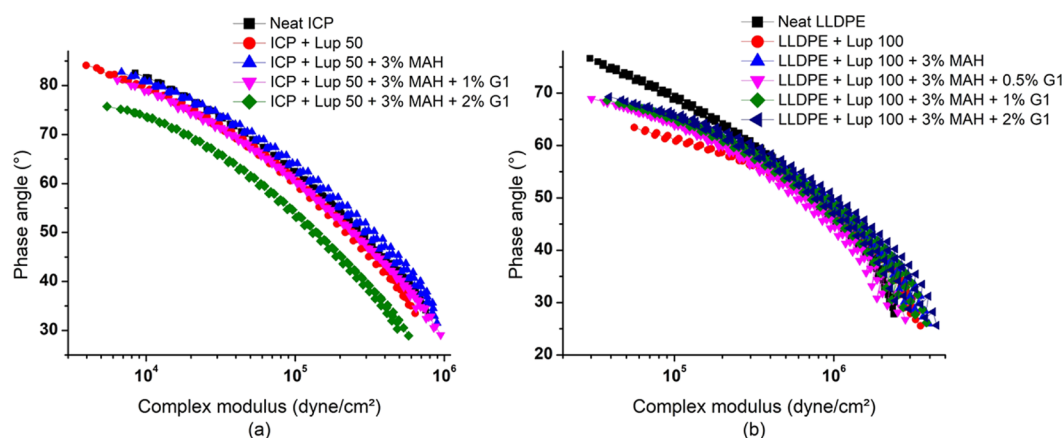


Figure 6. Van Gorp–Palmen plots for (a) neat ICP and ICP-G1 samples and (b) neat LLDPE and LLDPE-G1 samples.

reference temperatures of 190 °C for ICP and 160 °C for LLDPE.

Figure 5 shows complex viscosity versus angular frequency plots for ICP and LLDPE samples. In the case of ICP modification (Figure 5a), addition of Luperox resulted in severe chain scission, leading to a distinct decrease in the complex viscosity at a low frequency as well as broadening of the Newtonian zone. It increased with the grafting of MAH and further improved slightly with branching introduced with the grafting of 1% G1. The complex viscosity for modified ICP samples remained below neat ICP at low frequencies, which might be the reason for their improved processability and are also consistent with the values of zero shear rate viscosity listed in Table 7. For LLDPE modified with Luperox (Figure 5b), a huge increase in complex viscosity is a consequence of enhanced crosslinking between polymer chains. Complex viscosity increases with loading of G1 from 0.5% to 2% in LLDPE, indicating increased entanglement. An evident Newtonian plateau can be observed for neat LLDPE unlike modified LLDPE samples where shear thinning starts at low frequencies, which is a clear consequence of the introduction of long chain branching.^{20,45,46}

In the SAOS experiment, a “phase shift” between oscillatory deformation applied to the sample and the response generated by the material is measured in the form of a phase angle. Its value lies between 0° to 90° and gives a fundamental idea about the deformation of the material either elastically similar to a solid (implying a low phase angle) or inclined toward liquid-like viscous flow (implying a higher phase angle). Several literature reports have been published, showing the sensitivity of the phase angle toward long chain branching.^{47–50} Figure 6 shows the comparison of phase angles as a function of complex shear modulus, better known as the Van Gorp–Palmen (vGP) plot for neat and modified ICP and LLDPE samples. A decrease in the phase angle for both modified ICP and LLDPE samples corresponding to a particular value of complex shear modulus can be clearly observed. The vGP plot for neat ICP and LLDPE shows excellent superposition of linear viscoelastic data, indicating simple thermorheological behavior. A decrease in the phase angle and nonsuperposition of viscoelastic data are more evident in the case of modified LLDPE samples. Since the polydispersity index remained almost the same for all the LLDPE samples as mentioned previously, nonsuperposition can be attributed to the fact that long branched chains result in different temperature dependences of relaxation time, which

explains their thermorheological complexity.⁵¹ For better visualization, the curves were shifted because of overcrowding and are shown in Figure S10.

Another major parameter to detect branching through rheological data is by examining the flow activation energy, which can be considered as potential energy needed by the molecules for relaxation.^{52–54} The time–temperature superposition (TTS) of oscillatory shear data obtained at four different temperatures was shifted to a master curve and the shift factors were evaluated to determine the flow activation energy (E_a). The master-curve plots of storage and loss moduli as a function of angular frequency for all the samples are provided in the Supporting Information (Figures S11 and S12). ICP with 1% G1 grafted showed a distinctive increase in E_a (62.3 kJ/mol) as compared to that of neat ICP (35.5 kJ/mol). It also showed an increase in zero shear rate viscosity and relaxation time with respect to ICP melt blended with Luperox. ICP modified with 2% G1 is not following the trend, which means that addition of G1 beyond 1% with 50 ppm of Luperox is not desirable as it displays reduced storage modulus, loss modulus (Figure S11), and activation energy.

Rheological analysis was observed to be more pronounced for modified LLDPE samples. Zero shear rate viscosity drastically increased with addition of 100 ppm of Luperox due to enhanced crosslinking of LLDPE chains. It reduced with the grafting of G1 by decreasing excessive crosslinking between LLDPE chains brought by adding Luperox to LLDPE. It should be noted here that the zero shear rate viscosity of all modified samples was much higher than that of neat LLDPE. All the modified LLDPE samples showed higher values of storage modulus and loss modulus as compared to neat LLDPE. The breakthrough for detecting long chain branching in LLDPE was observed on comparing E_a values. While E_a for neat LLDPE was 16.6 kJ/mol, it increased up to 50.5, 58.6, and 68 kJ/mol with addition of 0.5, 1, and 2% G1, respectively. Since activation energy does not depend on the molecular weight or molecular weight distribution, an extensive increase in E_a can be attributed to reduced segmental dynamics with the introduction of long chain branching. A higher relaxation time was also observed for modified LLDPE samples owing to the simultaneous presence of long chain branched and linear chains in the polymer backbone.^{50,55–57}

CONCLUSIONS

The dendrimer G1 synthesized using TEMPO as an initiator for thiol-ene chemistry has proven to be a convenient approach in terms of yield, purification, ease, and time of reaction as compared to the conventional methods of synthesizing dendrimers. The reactive compatibilization carried out to graft G1 on the polyolefin backbone using MAH as a compatibilizer leads to incorporation of long chain branching in the impact copolymer PP and LLDPE chains. The presence of branching was confirmed using ^{13}C NMR, FT-IR, HT-GPC, and rheological parameters obtained through a parallel-plate rheometer.

Thermal, rheological, and HT-GPC analysis indicated that ICP modified with 1% G1 controlled the chain degradation to a large extent, which was brought about by addition of the radical initiator Luperox. It possessed improved processability or a higher MFI than neat ICP for the same magnitude of melt strength. It also had higher values of cross-over modulus and E_a than neat ICP. HT-GPC analysis of the rubber phase for the same composition of ICP showed a higher molecular weight and intrinsic viscosity as well broader molecular weight distribution as compared to neat ICP, which is a direct consequence of higher entanglement. The effect of long chain branching was more prominent for the modified LLDPE samples. The grafting of G1 controlled the excessive crosslinking as a result of addition of the radical initiator. All the compositions modified using G1 possessed higher melt strength, complex viscosity, storage as well as loss modulus, and zero shear rate viscosity with respect to neat LLDPE. The compositions with 0.5 and 1% additives had improved melt strength with similar MFI values as neat LLDPE. A huge increase in the values of flow activation energy for the modified LLDPE samples as compared to neat LLDPE was a clear indicator of higher entanglement and the presence of long chain branching in the polymer chains.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.iecr.1c01190>.

Characterization of the dendrimer G1, ^1H NMR, ^{13}C NMR, TGA, FT-IR, and FT-IR spectra, DSC thermogram, TGA curves, shifted Van Gurp–Palmen plots, and storage and loss modulus curves of neat and modified ICP and LLDPE samples (PDF)

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Notes

The authors declare no competing financial interest.

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